

A1_Code.py

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from matplotlib.animation import FuncAnimation
4
5 x_start = 0
6 x_end = 1
7
8 t_start = 0
9 t_end = 1
10
11 nx = 100 #number of grid points for x
12 CFL = 0.8
13
14 a = 1
15
16 dx = (x_end-x_start)/nx
17 dt = (CFL*dx)/a
18
19 nt = int((t_end-t_start)/dt)
20
21 x = np.arange(x_start,x_end,dx)
22
23 u0 = np.exp(-100*(x-0.5)**2)
24
25 u_n = np.zeros(len(x))
26 u_np1 = np.zeros(len(x))
27 u_history = np.zeros((nt,nx))
28
29
30 u_n = u0.copy()
31
32 for n in range(nt):
33     for i in range(nx):
34         if i == 0:
35             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[-1])
36         else:
37             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[i-1])
38     u_n = u_np1.copy()
39     u_history[n,:] = u_n.copy()
40
41
42 plt.plot(x,u_np1)
43 plt.xlabel("x")
44 plt.ylabel("u")
45 plt.title("t = 1")
46 plt.show()
47
48
```

```
49 plt.plot(x,u_history[0,:],"k",label="Initial Solution")
50 plt.plot(x,u_history[-1,:],"r",label="Final Solution")
51 plt.xlabel("x")
52 plt.ylabel("u")
53 plt.legend()
54 plt.title("Initial and Final Solutions")
55
56 plt.show()
57
58
59 fig, ax = plt.subplots()
60
61 line, = ax.plot(x, u_history[0, :])
62 ax.set_xlim(x_start, x_end)
63 ax.set_ylim(np.min(u_history), np.max(u_history))
64
65 def update(frame):
66     line.set_ydata(u_history[frame, :])
67     ax.set_title(f"Time = {frame * dt:.3f} s")
68     return line,
69
70 animation = FuncAnimation(
71     fig,
72     update,
73     frames=len(u_history),
74     interval=50
75 )
76
77 plt.show()
78
79 # %%
80
81
```